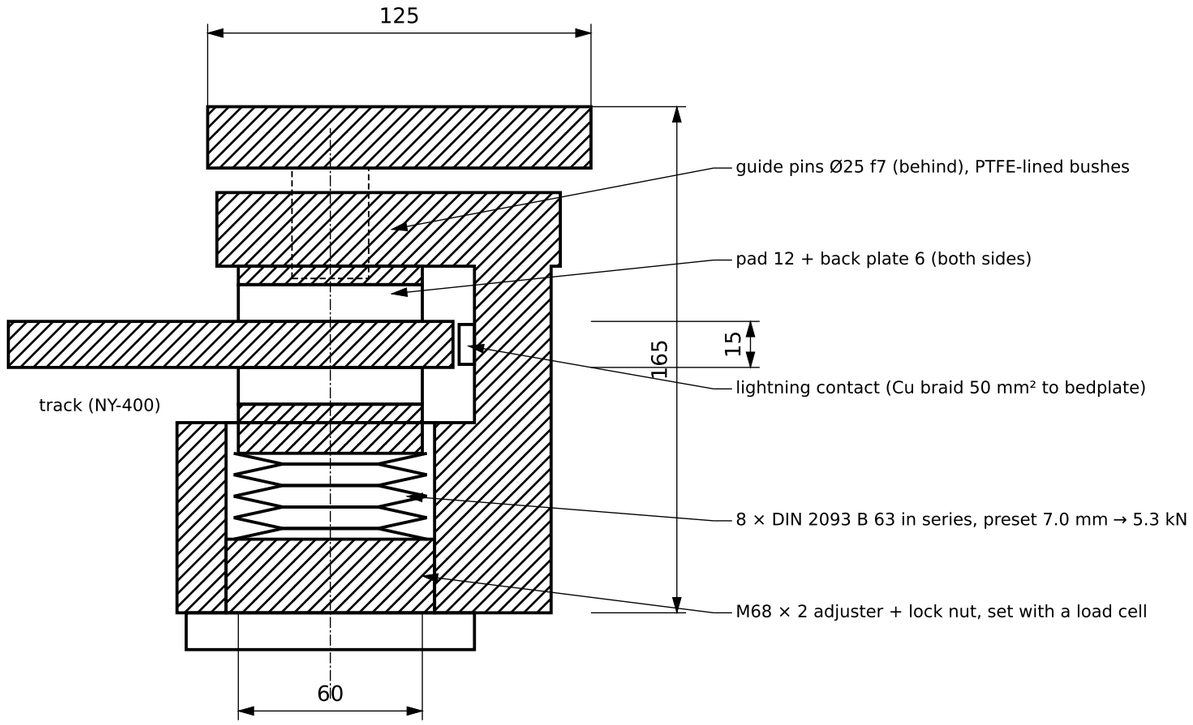
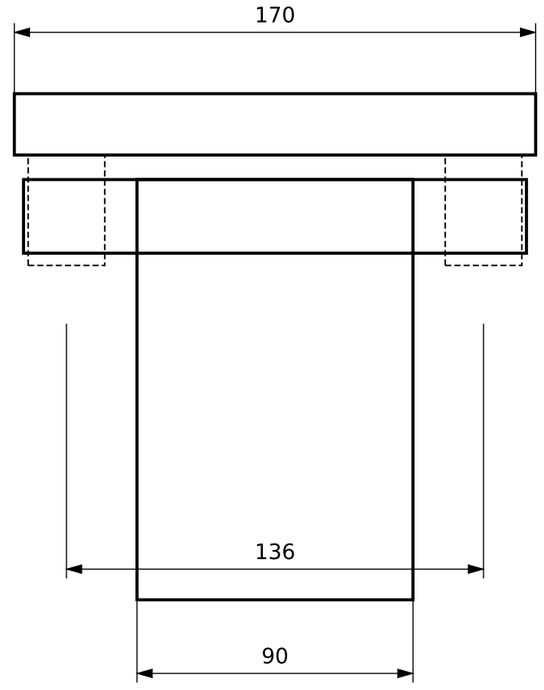
SECTION IN THE RADIAL PLANE · 1:2



VIEW FROM OUTSIDE (radial) · 1:2



Item	Part	Qty	Material / spec
1	Bracket 170 × 125 × 20, 4 × Ø17 to the bedplate	1	S355J2+N
2	C-frame (upper jaw, web, lower jaw), machined	1	S355J2+N
3	Guide pin Ø25 f7 × 55, pressed into item 1	2	1.4462 duplex
4	Pad 90 × 60 × 12 on 6 mm back plate	2	organic industrial pad, μ 0.40 ± 20 %
5	Piston Ø60 × 10	1	1.4462
6	Disc spring DIN 2093 B 63 (63 × 31 × 2.5)	8	51CrV4, zinc-flake
7	Adjuster M68 × 2 + lock nut	1+1	1.4462
8	Lightning contact block + Cu braid 50 mm²	1	CuSn / Cu

NOTES

1. Floating C-frame: the springs push the lower pad up; the reaction pulls the frame down onto the upper pad, so both faces clamp 5.3 kN at any pad wear.
2. Two units give 2.4 to 3.6 kNm (μ 0.32 to 0.48), 3.0 kNm nominal, 1.51 × the 95th percentile operating yaw moment.
3. Pad pressure 0.98 MPa, sliding speed 3.1 mm/s, heat 26 W. Estimated wear 6.6 mm in 20 years of 9 mm usable (to confirm by test).
4. Spring stack at 0.50 h0 per disc: one year of wear lowers the clamp by 8 %. Re-set at the yearly service (load cell between the pads).
5. Pad change without lifting the nacelle: back off the adjuster, swing the frame down on the pins, slide the pads out sideways.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · YAW SYSTEM (DOC 25)

FRICTION PAD UNIT (PASSIVE, SPRING-LOADED)

DRAWING NO.	REV / DATE	SCALE (A3)
NY-410	A · 2026-11-05	1:2
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
frame S355J2+N, pins 1.4462	13 kg (model)	Claude / not checked